

An Open, Reproducible Dashboard for Mapping the Canadian Metaresearch Landscape

The challenge — and our answer. The call asks: *“How can we construct an open, reproducible, and inclusive dataset that best captures the Canadian metaresearch landscape?”* Our answer is to define the corpus through a transparent, versioned query over OpenAlex (Priem et al. 2022) that anyone can re-run; to script every step from retrieval to visualization so the dataset rebuilds with a single command; to report several defensible definitions of “Canadian” and “metaresearch” side by side rather than hide a single choice; to co-design the inclusion criteria with the community; and to surface the dataset’s own gaps — language, discipline, coverage — directly in the dashboard rather than bury them. The deliverable is a live, public dashboard, already running as a pilot on real OpenAlex data, that lets researchers, funders, and communities explore who is shaping Canadian research culture, whose voices are missing, and where to improve transparency and equity.

Defining and retrieving the corpus. We operationalize “metaresearch” (Ioannidis et al. 2015) as a versioned OpenAlex query. The pilot draws on five OpenAlex topics — from OpenAlex’s current Topics taxonomy — , namely scientometrics and bibliometrics, academic publishing and open access, scholarly communication, academic integrity, and web visibility/informetrics, from 1970 onward (the field was named c.1969, so earlier topic tags are misclassifications) and excluding paratext, yielding roughly 7,000 Canadian-affiliated works; the full corpus will layer in English- and French-language title/abstract keywords. “Canadian” is defined by author–institution country in OpenAlex; because that choice is contestable, we report sensitivity checks against alternatives (first or corresponding author, funder) side by side. Retrieval uses the OpenAlex API with a dated, version-pinned snapshot, and we benchmark coverage against established sources such as Web of Science and Scopus (Mongeon and Paul-Hus 2016), validating relevance through a manually checked stratified sample and precision/recall against a curated seed set. Inclusion criteria are pre-registered (Nosek et al. 2018) and refined with stakeholders before the corpus is frozen. From the resulting dataset the dashboard derives output and citation trends, open-access patterns (including an open-versus-closed citation comparison (Piwowar et al. 2018)), thematic structure (topic and field breakdowns now, formal topic modeling for the full study), and co-authorship and institutional collaboration networks.

Openness, transparency, and reproducibility. Reproducibility is the project’s core design principle: the entire dataset rebuilds from source with one command, and every choice is inspectable. All extraction, cleaning, analysis, and dashboard code is MIT-licensed and public on GitHub; the processed, record-level corpus and its codebook are released under CC BY 4.0 and built to be FAIR (Wilkinson et al. 2016), with a dated OpenAlex snapshot so results remain reproducible as the database grows. The pipeline (raw → processed → dashboard) is fully scripted with pinned software environments, and the corpus-defining query lives in a single versioned configuration file, so the definition is auditable and re-runnable. Documentation includes a methodology and reproducibility guide, the codebook, and an in-dashboard “Search

& validation” panel that displays the exact query, re-runs it live against OpenAlex, and spot-checks a random sample for relevance — transparency any reader can exercise in the browser. The approach aligns with the UNESCO Recommendation on Open Science (UNESCO 2021).

Diversity across traditions, languages, and communities. Metaresearch is not monolithic, and the dataset is built to reflect that. Its traditions and disciplines span quantitative scientometrics, science-and-technology studies, research-integrity and ethics scholarship, and library and information science; our topic set deliberately mixes these, and the dashboard breaks the corpus down by field and topic so that no single tradition stands in for the whole. English and French are treated as first-class — the query, keywords, and dashboard are bilingual — and we report the English/French balance explicitly (the pilot is ~94% English, itself a finding rather than a target). Inclusion criteria are co-designed with librarians, researchers, EDI experts, and policymakers, and smaller institutions and all regions are tracked alongside the largest universities. Where the corpus touches Indigenous-led research or data, we follow the First Nations principles of OCAP® (First Nations Information Governance Centre (FNIGC), n.d.), the CARE Principles for Indigenous Data Governance (Carroll et al. 2020), and Chapter 9 of the Tri-Council Policy Statement (TCPS 2) (CIHR, NSERC, SSHRC 2022).

Assumptions, limitations, and sources of bias. We are explicit about what this approach assumes and where it can mislead. It assumes that OpenAlex topics and affiliations are usable proxies for “metaresearch” and “Canadian” — both contestable, which is why we test alternatives rather than assert one — and that affiliation country adequately captures Canadian contribution. The main limitations are coverage bias, since OpenAlex favours English-language, journal-based, DOI-bearing outputs and under-represents non-English work, grey literature, and community-based scholarship (Mongeon and Paul-Hus 2016; Vera-Baceta et al. 2019); definitional bias, since topic tagging is imperfect (broad topics can pull in tangential work) and affiliation-based attribution credits a whole work to Canada regardless of contribution share; and metadata gaps, since OpenAlex carries little reliable information on author demographics, career stage, or equity-deserving groups, so any equity analysis relies on coarse proxies. We surface these gaps in the dashboard rather than hide them, so the tool informs capacity building and equitable policy.

Work plan. All work is completed before the conference, in four overlapping phases:

Phase	Window	Output
Corpus definition, retrieval & stakeholder codesign	Jul–Aug 2026	Versioned query, dated snapshot, validated inclusion criteria
Cleaning, validation & bias assessment	Aug 2026	Validated dataset + bias/coverage report
Analyses & dashboard build	Sep–Oct 2026	Bibliometric, network, and topic results in a deployed dashboard
Documentation & rehearsal	Oct 2026	Reproducibility docs + 30-minute plenary talk

Outputs. An open dataset with codebook, reproducible code and workflows, a deployed public dashboard (CC BY 4.0 data and docs; MIT code), and a 30-minute plenary presentation.

References

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